

Problem 8.1

a) Coefficient of u_p must be 1.

b) $u_n + u_p$ is another.

c) always $u_n = 0$

Problem 8.2

$$U = \begin{bmatrix} 1 & 2 & 3 & 0 \\ 0 & 0 & 4 & 0 \end{bmatrix} \Rightarrow R = \begin{bmatrix} 1 & 2 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$\Rightarrow u = \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix}$$

$$U = \begin{bmatrix} 1 & 2 & 3 & 5 \\ 0 & 0 & 4 & 8 \end{bmatrix} \Rightarrow R = \begin{bmatrix} 1 & 2 & 0 & -\frac{1}{4} \\ 0 & 0 & 1 & 2 \end{bmatrix}$$

$$\Rightarrow u = \begin{bmatrix} -3 \\ 1 \\ 2 \end{bmatrix}$$

Problem 8.3

Yes.

$$Ay = b = Cy$$